

Advanced iOS Quiz

14 Questions | Multiple Choice + Complete the Code | Week 1-12 + Git

Student Name : _____

Score : / 14

1. In MVVM (iOS 13+), which property wrapper should a view use when it CREATES and OWNS the ViewModel for its entire lifetime?

- A. `@ObservedObject var vm: FarmViewModel`
- B. `@StateObject private var vm: FarmViewModel`
- C. `@EnvironmentObject var vm: FarmViewModel`
- D. `@State private var vm = FarmViewModel()`

2. Why should you use `NSPredicate` on a `@FetchRequest` instead of `Swift .filter{}` to show only expense transactions?

- A. `NSPredicate` is faster to write and has cleaner syntax
- B. `NSPredicate` filters inside SQLite, loading only matching rows into memory
- C. `Swift .filter{}` cannot compare `String` properties
- D. `.filter{}` requires a different `managedObjectContext`

3. What does the `@NSManaged` attribute tell the Swift compiler when applied to a `CoreData` property?

- A. The property is immutable after creation
- B. The property's storage is handled by `CoreData`'s Objective-C runtime, not Swift
- C. The property will be automatically saved every second
- D. The property is excluded from the `.xcdatamodeld` schema

4. In iOS 13+, what is the correct way to navigate programmatically to a detail view from code (not just from a tap)?

- A. `NavigationStack { }.navigationDestination(for: UUID.self)`
- B. `NavigationLink(tag: id, selection: $selectedID) { DetailView() } label: { EmptyView() }`
- C. `NavigationLink(destination: DetailView()) { Text("Go") }`
- D. `@Environment(\.dismiss) var dismiss`

5. Which class and method do you call first before scheduling a local notification in iOS 13+?

- A. `NotificationCenter.default.post(name:object:)`
- B. `UNUserNotificationCenter.current().requestAuthorization(options:completionHandler:)`
- C. `UNNotificationRequest.schedule(after:repeats:)`
- D. `UserDefaults.standard.set(true, forKey: "notificationsAllowed")`

6. What does `@ViewBuilder` enable in a generic container like `DashboardSection<Content: View>?`

- A. It allows the struct to inherit from `UIView`
 - B. It lets the caller pass a closure returning multiple SwiftUI views without wrapping in a `Group`
 - C. It enables live-preview rendering in Xcode Canvas
 - D. It replaces the need for `@EnvironmentObject` in child views
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7. What is the primary advantage of extracting repeated `.padding().background().cornerRadius()` into a custom `ViewModifier`?

- A. `ViewModifiers` run on a background thread and improve performance
- B. Styling is centralised -- changing the modifier updates every view that applies it
- C. `ViewModifiers` bypass the SwiftUI layout system for faster rendering
- D. `ViewModifiers` allow you to call UIKit methods directly

8. In iOS 13+, which API is used to generate a formatted PDF report with text and tables?

- A. `ImageRenderer` -- capture a SwiftUI view as a PDF
- B. `UIGraphicsPDFRenderer` -- draw text, tables, and graphics into a PDF context
- C. `PDFDocument` from `PDFKit` with `appendPage()`
- D. `ShareLink` with a URL to a pre-built template

9. (Complete the Code) What belongs on the blank line to persist the record across app restarts?

```
func addTransaction(
    amount: Double, note: String,
    type: String, category: String
) {
    let t = Transaction(context: context)
    t.amount = amount
    t.note = note
    t.type = type
    t.category = category
    t.id = UUID()
    _____ // <- what goes here?
}
```

- A. `context.insert(t)`
- B. `context.refresh(t, mergeChanges: true)`
- C. `saveContext()`
- D. `context.fetch(Transaction.fetchRequest())`

10. (Complete the Code) What should picker.delegate be set to?

```
struct ImagePicker: UIViewControllerRepresentable {
    @Binding var image: UIImage?
    func makeCoordinator() -> Coordinator { Coordinator(self) }
    func makeUIViewController(context: Context)
        -> UIImagePickerController {
        let picker = UIImagePickerController()
        picker.delegate = _____
        return picker
    }
}
```

- A. self
- B. context.coordinator
- C. UIImagePickerController()
- D. Coordinator(self)

11. Before running Product -> Archive in Xcode to upload to TestFlight, what must you select in the device/scheme destination picker?

- A. iPhone 15 Pro (Simulator)
- B. My Mac (Designed for iPad)
- C. Any iOS Device (arm64)
- D. Generic iOS Device (x86_64)

12. Why must LaunchScreen.storyboard contain only static images -- no Swift code, no animations, no @IBOutlet connections?

- A. Xcode strips all code from storyboard files during archive
- B. The OS renders the launch screen before the app process starts -- no Swift runtime is available yet
- C. Animations in storyboards require iOS 16+ and break backward compatibility
- D. App Store review tools flag storyboards that reference Swift classes

13. What is the correct order of Git's three storage areas when you save a code change permanently?

- A. Repository -> Staging Area -> Working Directory
- B. Working Directory -> Repository -> Staging Area
- C. Staging Area -> Working Directory -> Repository
- D. Working Directory -> Staging Area -> Repository

14. Which single Git command creates a new branch 'feature/login' AND immediately switches to it?

- A. git branch feature/login
 - B. git merge feature/login
 - C. git switch -c feature/login
 - D. git push -u origin feature/login
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